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Parameter personalisasi : $N = 23$, $a = 2$, $b = 3$, $k = 6$, $\theta_0 = 75°$, $d = 0{,}004$

Soal 3

a. Definisi Himpunan Pelanggan

$A = \{23, 25, 27, 30, 33, 36, 38\}$ ← pelanggan dari medsos

$B = \{26, 28, 30, 33, 35, 41\}$ ← Pelanggan dari referal

$C = \{24, 30, 33, 34, 40, 43\}$ ← Pelanggan dari loyalty program

$U = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43\}$

1. $A \cup B \cup C = \{23, 24, 25, 26, 27, 28, 30, 33, 34, 35, 36, 38, 40, 41, 43\}$

   $|A \cup B \cup C| = 15$

2. $A \cap B \cap C = \{30, 33\}$

   $|A \cap B \cap C| = 2$

3. $(A \cup B) \cap C' =$

   $A \cup B = \{23, 25, 26, 27, 28, 30, 33, 35, 36, 38, 41\}$

   $C' = U - C = \{23, 25, 26, 27, 28, 29, 31, 32, 35, 36, 37, 38, 39, 41, 42\}$

   $(A \cup B) \cap C' = \{23, 25, 26, 27, 28, 35, 36, 38, 41\} = 9$

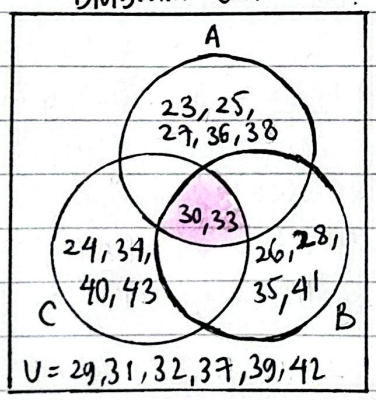
4. $A \oplus B = (A \cup B) - (A \cap B)$

   $A \cap B = \{30, 33\}$

   $A \cup B = \{23, 25, 26, 27, 28, 30, 33, 35, 36, 38, 41\}$

   $A \oplus B = \{23, 25, 26, 27, 28, 35, 36, 38, 41\} = 9$

Diagram Venn



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c. Fungsi Harga

Diket : $f(x) = kx + (a+b)$

$g(x) = \frac{x^2}{b+1} + a$

Jawab :

$f(x) = 6x+(2+3)$ $\quad g(x) = \frac{x^2}{3+1} + 2 = \frac{x^2}{4} + 2$

$= 6x + 5$

1. $(f \circ g)(x) \rightarrow f(g(x)) = 6\left(\frac{x^2}{4}+2\right)+5$

$= \frac{6x^2}{4} + 12 + 5$

$= \frac{3}{2}x^2 + 17$

2. $(g \circ f)(x) \rightarrow g(f(x)) = \frac{6x+5^2}{4} + 2$

$= \frac{36x^2+60x+25}{4} + 2$

$= 9x^2 + 15x + \frac{25}{4} + 2$

$= 9x^2 + 15x + \frac{33}{4}$

Uji Komutatif

$x = b + 2$

$= 3 + 2$

$= 5$

$f(g(5)) \rightarrow g(5) = \frac{25}{4} + 2 = \frac{33}{4}$

$f = 6 \cdot \frac{33}{4} + 5$

$= \frac{198}{4} + 5$

$= \frac{218}{4} = 54{,}5$

$g(f(5)) \rightarrow f(5) = 6(5) + 5 = 35$

$g = \frac{35^2}{4} + 2$

$= \frac{1225}{4} + 2$

$= \frac{1233}{4} = 308{,}25$

Kesimpulan : $f(g(x)) \neq g(f(x)) \rightarrow$ tidak komutatif.

d. Invers $f(x)$

diket : $y = 6x+5$

tukar : $y = 6x+5$

$x = \frac{y-5}{6}$

$f^{-1}(x) = \frac{x-5}{6}$

$y = k \cdot 100 + a + b$

$= 6 \cdot 100 + 2 + 3$

$= 600 + 5$

$= 605$

$x = \frac{605-5}{6} = \frac{600}{6} = 100$

$f(100) = 600 + 5 = 605$ (benar)

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e. Klasifikasi

Diket : $f(x) = 6x+5$

* Uji Injektif (one to one) $\rightarrow f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

$f(x_1) = f(x_2)$

$6x_1 + 5 = 6x_2 + 5$

$6x_1 = 6x_2$

$x_1 = x_2$ (terbukti benar)

* Uji Surjektif (onto) $\rightarrow y \in R$, ada $x \in R \rightarrow f(x) = y$

$y = 6x+5$

$x = \frac{y-5}{6}$

karena $x \in R$, maka setiap y selalu ada pasangan x. (terbukti benar)

Diket : $g(x) = \frac{x^2}{4} + 2$

* Uji Injektif

$g(2) = \frac{4}{4} + 2 = 3$

$g(2) = g(-2)$ tetapi $2 \neq -2$ (tidak memenuhi)

* Uji Surjektif

$g(x) = \frac{x^2}{4} + 2 \geq 2$

artinya, hasil fungsi tidak pernah < 2

jika $y = 1$, tidak ada yg memenuhi $\frac{x^2}{4} + 2 = 1$, jadi tidak semua elemen kodomain tercapai.

- Fungsi $f(x) = 6x+5$ terbukti injektif karena memenuhi definisi $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Dan surjektif karena utk setiap $y \in R$ terdapat $x = \frac{y-5}{6}$. Oleh karena itu, f adalah fungsi bijektif.

- Fungsi $g(x) = \frac{x^2}{4} + 2$ tidak injektif karena $g(2) = g(-2)$ tetapi $2 \neq 2$, serta tidak surjektif karena hasil fungsi selalu ≥ 2. Sehingga tidak mencakup seluruh R. Maka, g bukan bijektif.

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